

Web Science

Assignment – 3



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1 Recursive Query : DNS

You have solved the "Routing Table" question in Assignment 2. We updated the routing tables once more. resulting in the following tables creating the following topology

Table 1: Routing Table

Router1			Router2			Router3		
Destination	Next Hop	Interface	Destination	Next Hop	Interface	Destination	Next Hop	Interface
67.0.0.0	67.68.3.1	eth 0	205.30.7.0	205.30.7.1	eth 0	205.30.7.0	205.30.7.2	eth 0
62.0.0.0	62.4.31.7	eth 1	156.3.0.0	156.3.0.6	eth 1	88.0.0.0	88.6.32.1	eth 1
88.0.0.0	88.4.32.6	eth 2	26.0.0.0	26.3.2.1	eth 2	25.0.0.0	25.03.1.2	eth 2
141.71.0.0	141.71.20.1	eth 3	141.71.0.0	141.71.26.3	eth 3	121.0.0.0	121.0.3.1	eth 3
26.0.0.0	141.71.26.3	eth3	67.0.0.0	141.71.20.1	eth 3	156.3.0.0	205.30.7.1	eth 0
156.3.0.0	88.6.32.1	eth 2	62.0.0.0	141.71.20.1	eth 3	26.0.0.0	205.30.7.1	eth 0
205.30.7.0	141.71.26.3	eth 3	88.0.0.0	141.71.20.1	eth 3	141.71.0.0	205.30.7.1	eth 0
25.0.0.0	88.6.32.1	eth 2	25.0.0.0	205.30.7.2	eth 0	67.0.0.0	88.4.32.6	eth 1
121.0.0.0	88.6.32.1	eth 2	121.0.0.0	205.30.7.2	eth 0	62.0.0.0	88.4.32.6	eth 1

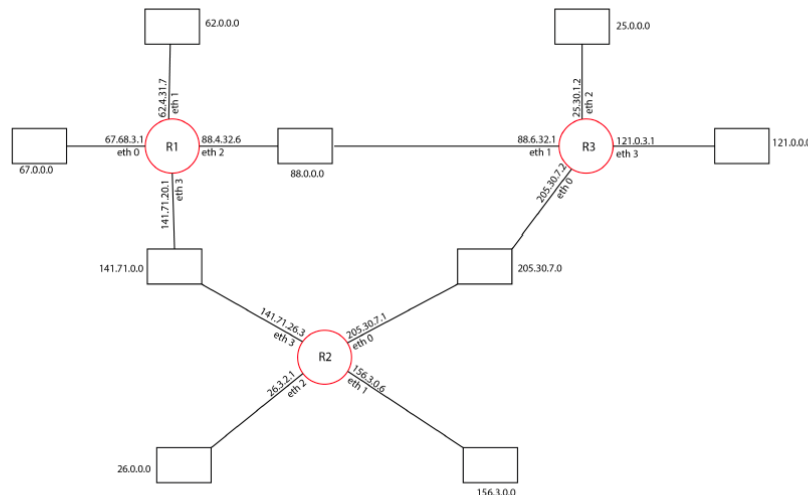


Figure 1: DNS Routing Network

Let us assume a client with the following ip address 67.4.5.2 wants to resolve the following domain subdomain.websciencexampldomain.com using the DNS.

You can further assume the root name server has the IP address of 25.8.2.1 and the name-server for websciencexampldomain.com has the IP address 156.3.20.2. Finally the sub-domain is handled by a name server with the IP of 26.155.36.7.

Please explain how the traffic flows through the network in order to resolve the recursive DNS query. You can assume ARP tables are cached so that no ARP-requests have to be made.

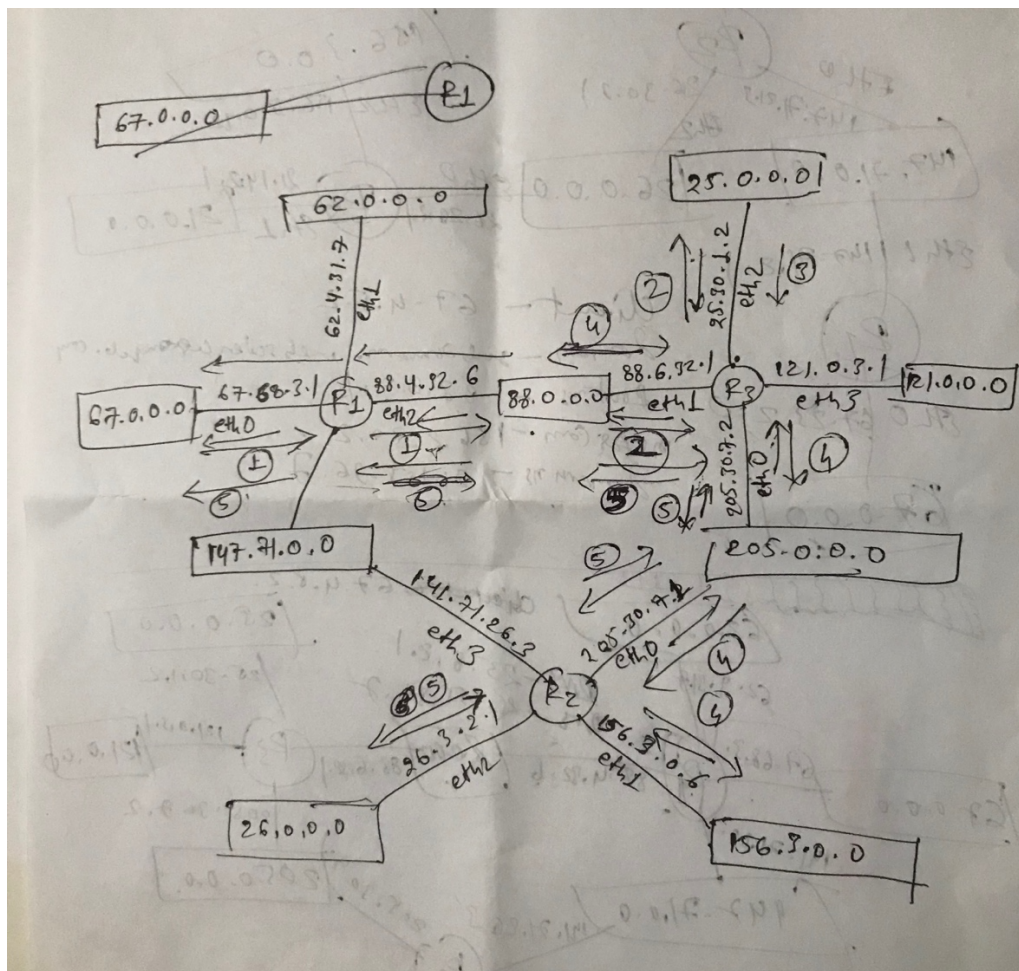
Answer – The client (67.4.5.2) requests dns server to 67.68.3.1 to R1. Similarly, R1 to 88.4.32.6 to 88.0.0.0. Then, the ISP dns server name server sets a request to the root name server.

So, 88.0.0.0 requests to 88.6.32.1 to R3 and from R3 to 25.30.1.2 to 25.8.2.1. Now, root name server request IPS dns server to the .com name server, from 25.8.2.1 to 88.0.0.0, same as last path, but inverse.

After that, ISP dns will requests .com name server from 88.0.0.0 to 88.6.32.1 to R3 to 205.30.7.2 to 205.30.7.1 to R2 to 156.3.0.6 to 156.3.20.2. The .com server will set ISP dns server to the domain name and gives feedback to the root by same route backwards.

Now, The ISP dns server requests domain name server from 88.0.0.0 to 88.6.32.1 to R3 and from R3 to 205.30.7.2 to 205.30.7.1 to R2 to 26.3.2.1 to 26.155.36.7.

The domain name server will return the ip address of the url to the IPS dns server via same route backwards. Now that IPS dns server gets the ip address, it will send it to the client (67.4.5.2). It sends address to client via the same route as client to ISP dns server, but backwards.



3 Internet Architecture

- Summarize the 4 layers of Internet architecture

Answer – The four layers of Internet architecture are:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Link Layer

Application Layer contains all programs such as web browsers and web servers that exchange information or interact with other applications. Its job is to send files and information to the transport layer using the appropriate protocol. Example of this layer is using SMTP protocol to send emails, using FTP to transfer files, using HTTP for web- browsing,

Transport Layer determines how the information sent by the application layer is going to get to its desired locations. As all information can't be sent at once because of the big size, the transport layer breaks them into smaller parts called packets. Then the port is decided to send those packets to the internet layer. And also, if the data are lost during the transmission, it handles by requesting the data again.

Internet layer deals with IP addresses. It adds source and destination IP addresses and sends packets received from the transport layer to the destination irrespective of the network or the route it takes to reach the destination address.

Link layer is the physical connection between network devices. It adds the MAC address of the destination and is sent. These MAC addresses are changed over the period of time on every hop till it reach its destination. Example of this is Ethernet or WIFI.

- Provide a single example that shows the presence and significance of each of the aforementioned 4 layers.

Answer – When we send some text using the web browser, HTTP protocol is used by the application layer. When the application receives data to send, it sends the data to the Transport layer where the port is decided to transfer the data reliably to the destination. To deliver the data to the destination, HTTP uses port 80. Then the transport layer breaks the data into packets and transfers those packets using a port to the Internet layer which adds the source and destination IP address and passes it to the Link layer. Finally, the link layer delivers the packets using a physical connection like WIFI where data are hop form one to another until it reaches the destination.

